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Development of a Computer Vision-based Football Video Analytics Pipeline

Computer Vision Pipeline for Football Analytics

This project presents an offline, batch-processing computer vision pipeline designed for comprehensive football match footage analysis, generating player-level, team-level, and ball-related analytics.

- **Detection and Tracking:** Utilizes YOLO-based detection combined with ByteTrack for robust multi-object tracking of players and referees.
- **Specialized Modules:** Integrates specialized components for accurate team classification and high-resolution ball detection throughout the footage.
- **Analytics Extraction:** Successfully extracts meaningful spatial and temporal information from raw broadcast video.
- **Evaluation:** Performance validated on 30-second clips from the SoccerNet dataset to ensure high-fidelity analytical outputs.

Addressing the ‘Data Gap’: Moving Beyond Spatial Tracking

Current tracking data is **inherently fragile**, with occlusions causing track fragmentation and ID switches. Single-frame snapshots provide no temporal context, making it difficult to differentiate tactical passes from routine dribbles.

Key challenges and research focus:

- **Limitations of Existing Datasets:** SoccerNet-v2 relies on temporally sparse 'Action Spotting' labels which lack the density for complex event modeling.
- **Structured Temporal Intelligence:** Moving beyond simple spatial coordinates to engineer context-aware features.
- **Advanced Feature Derivation:** Extracting physics-based metrics and high-level team analytics to bridge the data gap.

System Validation: SoccerNet Dataset Pipeline

The pipeline is validated using the SoccerNet dataset, which captures the dynamic nature of broadcast environments, including camera motion, varying illumination, and frequent occlusions.

Spatial Resolution: 1280x720 (720p)

- Native resolution balances visual feature distinctness with computational load.

Segment Duration: 30-Second Clips

- Captures full tactical transitions without the memory overhead of processing full 90-minute matches.
- System processing effectively manages the complex visual artifacts inherent in professional sports broadcasting.

Manifest-Based Architecture & Data Decoupling

System Design

The project implements a 'manifest-based' architecture, strictly decoupling execution logic from the underlying data. All experiments are driven by a centralized CSV manifest file.

This design promotes experiment auditability and scalability, allowing researchers to modify the analytical scope by updating the manifest without altering the core pipeline.

Manifest Record Fields

Each record in the centralized CSV defines the following parameters:

- Clip identifier
- Relative path
- Data split (train/val/test)
- Frame rate
- Resolution

System Architecture: Four-Phase Processing Pipeline

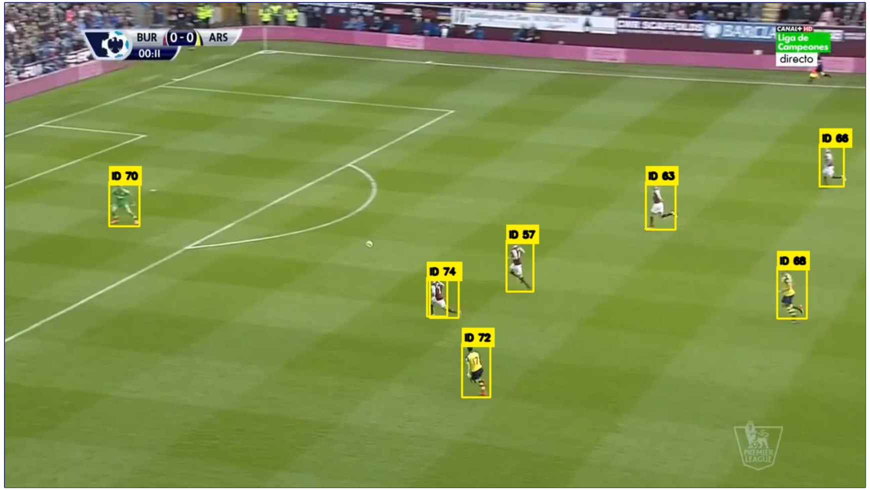
The system architecture operates in four distinct phases to transform raw video data into predictive tactical insights:

1. **Vision Extraction:** YOLOv8 and ByteTrack convert raw video into 2D bounding boxes.
2. **Temporal Module:** Processes raw tracks into context-aware physics and spatial metrics.
3. **Heuristic-Based Weak Labeling:** Interprets engineered features to label discrete tactical states (Carries, Passes, Interceptions).
4. **Deep Learning Preparation:** Aggregates telemetry into fixed-length temporal windows (e.g., 64-frame sequences) for training a Multiclass BiGRU network.

Stage 1: Player Localization with YOLOv8n

The first stage localizes all visible players using a lightweight **YOLOv8n** detector. Each frame is processed independently, outputting bounding boxes, class labels, and confidence scores.

- Detections are filtered to keep only the **'person'** class.
- Operates at a **640-pixel** input size for balance between precision and speed.
- Outputs are exported as **CSV files** containing frame number, timestamp, coordinates, confidence, and class label.



YOLOv8n Real-time Player Localization Output

ByteTrack: Multi-Object Tracking & Association Strategy

ByteTrack is used to associate detections across consecutive frames and assign persistent track identifiers, significantly reducing track fragmentation during occlusions or motion blur.

Two-Stage Association Strategy:

- **First Stage:** Match high-confidence detections to existing tracks to maintain reliable trajectories.
- **Second Stage:** Recover lower-confidence detections by matching them with remaining tracklets, handling partial occlusions.

Output Data Structure (CSV Format):

- Persistent Track ID for each player
- Frame Number and Timestamp
- Bounding Box Coordinates [x, y, w, h]
- Calculated Center Point

Team Classification and Feature Extraction

Visual Evidence Collection

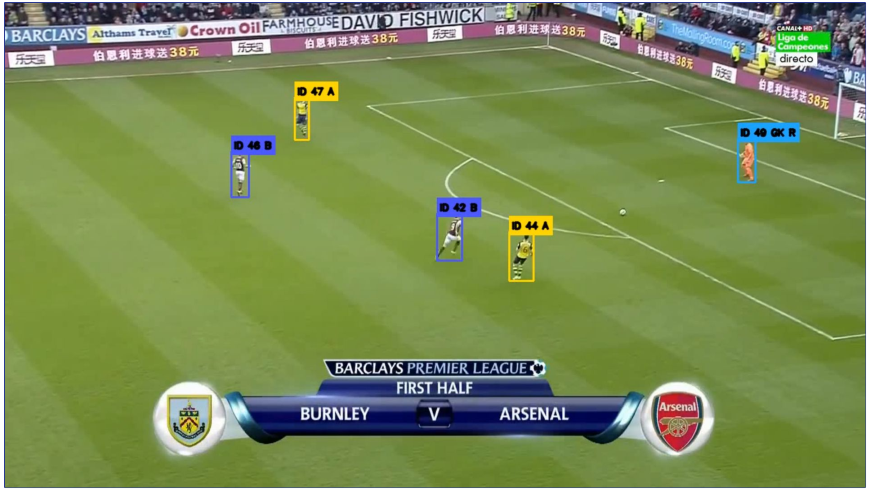
Team classification is track-based, collecting visual evidence from several samples of the same track.

Preprocessing Pipeline

- Torso crops are extracted from tracked players.
- Pitch-green background pixels are removed via masking.
- Remaining pixels converted to LAB color space for lighting invariance.

Team Separation

K-Means clustering is applied to separate the two main outfield teams (Team A and Team B). Role-aware heuristics identify special cases such as goalkeepers and referees.



Automated Team Assignment Visualization: K-Means Team Separation

Ball Detection: High-Resolution YOLO Pipeline

Dedicated Detector Architecture

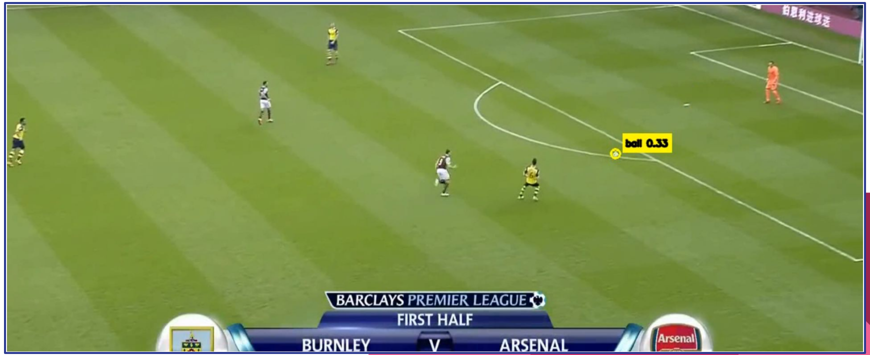
Uses a custom YOLO-based detector at 1280px resolution to preserve small object details like the football across wide pitch views.

Multi-Stage Filtering Logic

- **Geometric Constraints:** Filters by bounding box area, width, and height ratios.
- **Spatial Exclusion:** Ignores detections in the top broadcast graphics region.
- **Confidence Arbitration:** Retains only the highest-confidence candidate if multiple balls are detected.

The module exports both raw and filtered streams for full pipeline transparency and debugging.

Raw YOLO Detections (High Recall)



Filtered Detections (High Precision)

Ball Analytics: Quantitative Filtering & Tracking Results

High-Recall Baseline

The initial YOLO pass detected **1,233** raw candidate ball locations, introducing significant false-positive noise from advertising boards, spectators, and field graphics.

Precision Filtering Efficacy

Filters restricted data to **613** reliable detections—achieving a **49.7%** retention rate while eliminating major visual artifacts.

Track Reconstruction

Validated detections reconstructed into **27 unique ball tracks**. The tracking engine utilized linear interpolation to fill 82 gaps across 195 interpolated positions, yielding a consistent trajectory.

27
BALL TRACKS

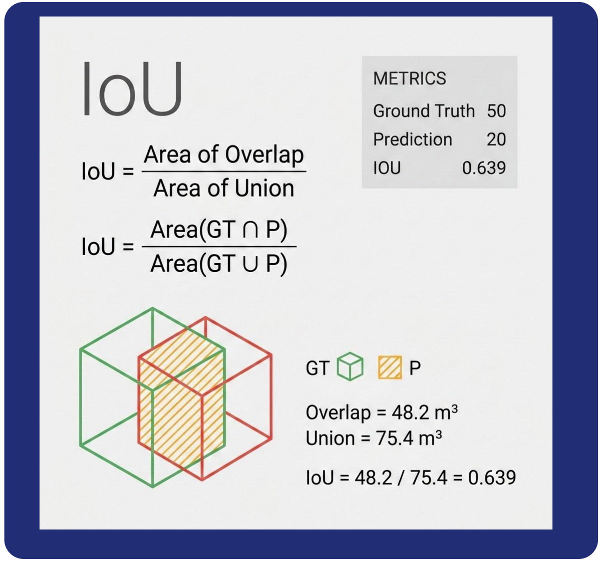
82
GAP FILLS

195
INTERP. POS.

System Validation: Detection & Spatial Accuracy

Spatial Evaluation Methodology

- **Bounding-Box Matching Policy:** Object localization quality is measured objectively by converting normalized annotation coordinates into pixel space and calculating `Intersection over Union (IoU)` fractions.
- **Greedy Matching Execution:** Predicted tracking boxes for players, referees, and the ball are greedily matched to unmatched CVAT/YOLO ground-truth boxes on a per-frame and per-class basis.
- **True Positive Criterion:** Detections are strictly classified as True Positives (TP) only when the IoU threshold satisfies $\geq$ `0.5`; scores below this are penalized as false positives.



Spatial Detection: Achieved high reliability across core tracked entities (Player F1: **0.9980**, Ball F1: **0.9811**).

Event Extraction: Heuristic Candidate Generation

The system infers actions from interactions between player/ball tracks and possession states.

Heuristic Core

- Generates candidates using heuristic rules rather than manual labels.
- Confidence scores based on possession stability and temporal duration.
- Calculated from ball-player proximity and team consistency metrics.

Event Types

- **Carries:** Sustained possession by a single player moving with the ball.
- **Passes:** Transfers of the ball between teammates.
- **Interceptions:** Disruption and capture of the ball from the opposing team.

Focus is placed on automated event identification through spatiotemporal feature analysis.

Player Movement Analytics: Trajectory and Tracking Summaries

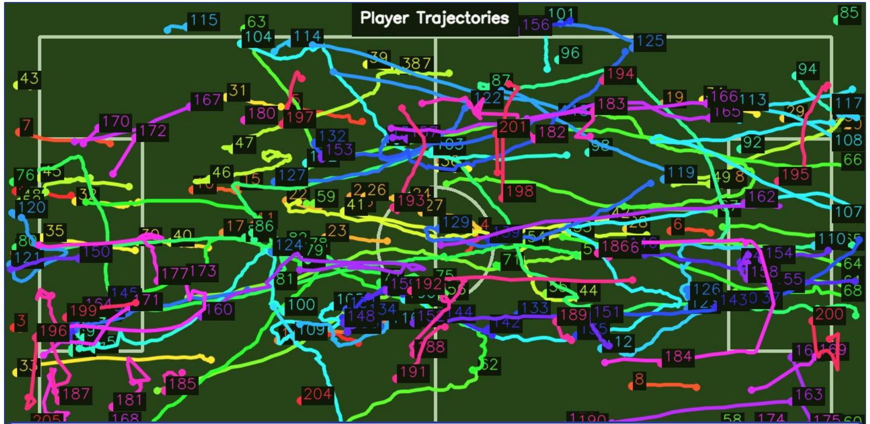
The system generates visual and numerical summaries describing player movement across the pitch.

Key Visualizations:

- **Trajectory Coverage:** Spatial visualizations show the cumulative coverage area of tracked players.
- **Active Player Plots:** Simplified trajectory plots highlight the most significant movements to improve readability.

Extracted Movement Statistics:

- Frames observed per individual track
- Mean confidence of the tracking engine
- Total distance traveled (px)
- Average player speed (px/s)

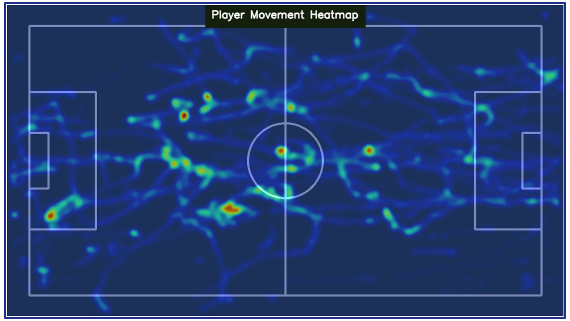


Numerical Track Summary

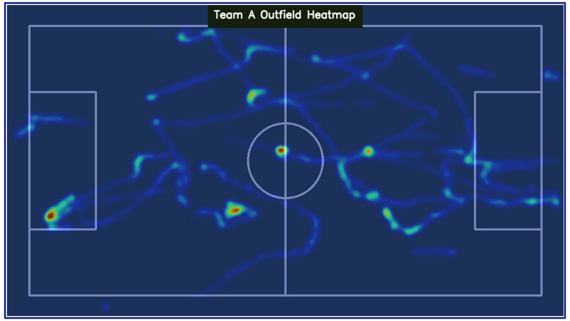
Automated feature extraction provides quantitative data for every tracked entity, facilitating downstream tactical analysis and performance benchmarking.

Qualitative Spatial Analysis: Team Movement Heatmaps

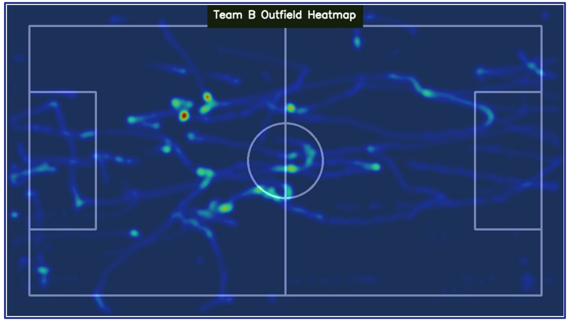
Player positions are accumulated into 2D density maps, providing a qualitative representation of spatial organization and movement tendencies for both teams.



Global Movement Heatmap (All Players)



Team A Heatmap (Outfield Assignments)



Team B Heatmap (Outfield Assignments)

Visualizing spatial density highlights tactical organization differences.

Possession Analytics: Statistical Integrity

Explicit Unlabeled States

Out of 778 frames with valid ball observations:
• **Team B:** 194 frames
• **Team A:** 50 frames
• **No Possession:** 534 frames (68%)
Confidence-based rejection of ambiguous tracking inputs.

Conservative Inference Strategy

Leaving loose balls or aerial passes unassigned acts as an **engineered smoothing block**. This prevents noise from corrupting downstream event-classification features.

Measured Spatial Tolerance

Average player-ball proximity: **73.73 px** during ownership.
Strict maximum cutoff: **146.91 px**.
Framework is driven exclusively by strict geometric evidence.

Integrity-first ownership logs ensure high-fidelity inputs for sequence modeling.

Dynamic Possession Estimation & Diagnostic Visualization

Ball Ownership Logic

- Estimated frame-by-frame by associating ball with the **nearest eligible player**.
- Diagnostic frames visualize selected player, ball confidence, and distance.
- **Low confidence** frames (below threshold) result in no possession assignment.

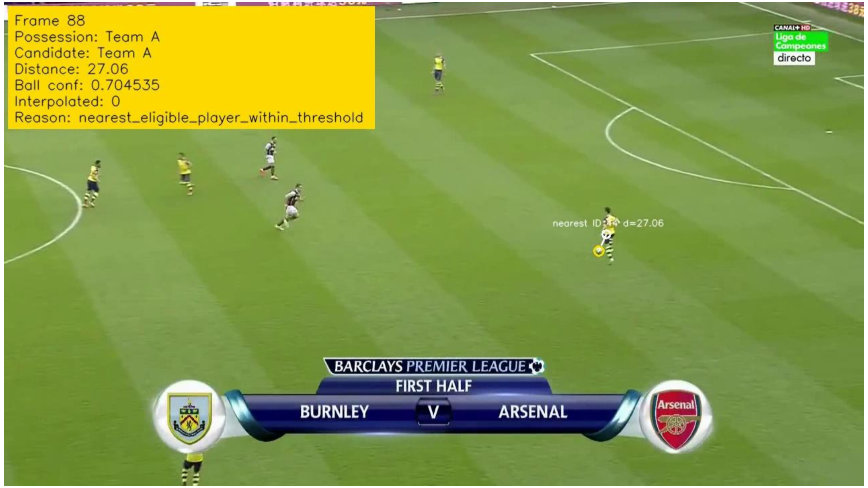


Fig 1: Diagnostic Frame - Nearest player association (d=27.06 px)

Stats: 73.73 px Avg Distance | Confirmed Possession Switches

Event Inference Methodology: Carry and Pass Candidates

Carry Candidate Generation

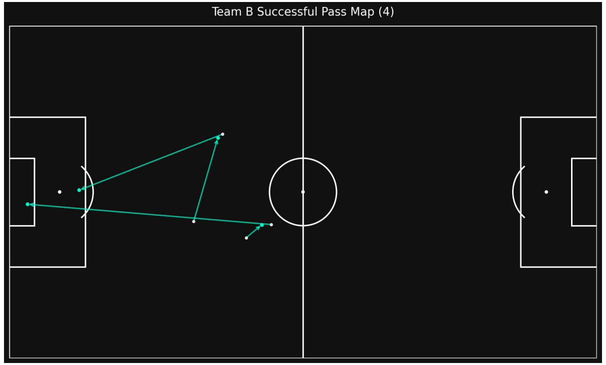
Retained possession during movement over stable time segments.

- **Possession duration**
- **Player displacement**
- **Mean ball-player distance**

Pass Candidate Inference

Possession transfers between teammates after stable segments.

- **Transit duration & distance**
- **Receiver-ball proximity**



Example: Inferred successful pass map visualization

Ball-player proximity tracking for candidate validation

Interception Candidate Identification and Visual QA

Derivation Logic

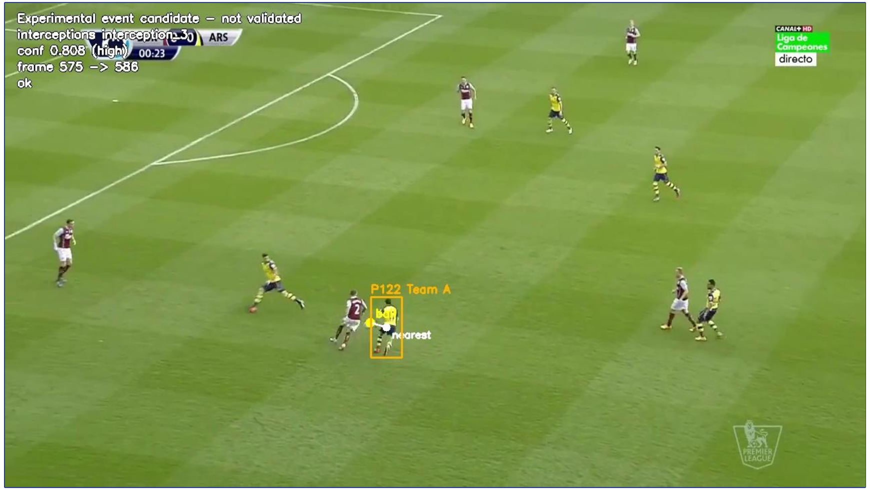
Interception candidates are derived from possession switches between opposing teams. Key predictive features include:

- Previous and new possession stability
- Team-switch confidence metrics
- Ball proximity to the new owner

Qualitative Inspection

Visual QA snapshots are generated for top candidates to ensure detection robustness, overlaying critical metadata:

- Event type and candidate ID
- Confidence scores and warning flags
- Ball position and involved players



Visual QA Snapshot: Interception candidate frame featuring automated tracking overlays and confidence scoring (high conf: 0.808).

Automated QA validates possession switch stability and team-switch confidence.

Supervision Transition: From Weak to Gold Labels

Weak Supervision Phase

Initial training utilized provisional labels generated from heuristic event candidates.

- A pilot experiment with ~200 heuristic windows proved feature learnability, justifying the large-scale manual annotation effort.

Gold-Standard Creation

Manual temporal annotation via CVAT on 25 match clips established ground truth.

- **2150 Total Windows**
- 1720 training samples
- 430 validation samples

CVAT Annotation Interface: Manual Temporal Boundary Identification

Canonical dataset ensures high-fidelity temporal action recognition.

Temporal Module: Engineering 68 Frame-Level Features

The temporal module aligns tracks and estimates into single structured records, stacking frame-level vectors into temporal windows for sequence processing.

Ball-State & Possession Variables

- Velocity and acceleration metrics
- Distance to ball possessor
- Possession change flags

Team-Shape Features

- Field width and depth metrics
- Spatial spread and compactness
- Estimated team centroids

Local Player-Context

- Proximity of nearest players
- Velocity vectors of nearest players
- Pressure and support metrics

Structured Record Output

- Total of 68 engineered numeric features
- Time-series stacking for windowing
- Unified track and estimate alignment

Model Evaluation: BiGRU vs. CNN-LSTM Architectures

BiGRU: Structured Sequence Model

- **Input:** 64-frame sequences of 68 engineered features.
- **Hidden Size:** 96 units.
- **Objective:** Evaluate the predictive value of structured analytics and engineered form metrics.

CNN-LSTM: Visual Temporal Model

- **Input:** 16 sampled RGB frames (resized to 112x112).
- **Feature Learning:** Learns visual embeddings directly from image content.
- **Objective:** Test if visual temporal information alone is sufficient for outcome prediction.

Research Focus: Comparing analytic feature value vs. raw visual sequences.

Temporal Model Performance Evaluation

Temporal Modeling Analysis:

- **Baseline BiGRU (Weakly Supervised):** A feasibility experiment testing if 68 temporal features contain useful info. Stopped at 18 epochs with a validation Macro F1 of 0.4660.
- **Weighted Focal-Loss BiGRU:** Addressed class imbalance using focal loss (γ=2.0) and inverse-frequency weights to penalize misclassification of rare events like turnovers.
- **CNN-LSTM:** Achieved Accuracy 0.5814 and Macro F1 0.2424. Excelled on frequent classes but struggled with rare events and tactical extraction from raw pixels.
- **BiGRU (Gold-label):** Highest Macro F1 at 0.2689 (0.4953 Accuracy), demonstrating superior effectiveness in identifying critical rare events.

The BiGRU architecture demonstrates the utility of structured engineered features for rare event detection in imbalanced datasets.

System Limitations & Constraints

2D Image-Space

The pipeline operates strictly within 2D video coordinates. Metrics like distance and speed are relative to camera angles and lens distortions, not absolute physical meters.

Class Imbalance

Temporal models are constrained by limited gold-standard training data. Rare events (carries, turnovers, shots) are underrepresented compared to passes.

Prototype Status

The system remains a proof-of-concept prototype. Requires broader manual annotation, scaling, and full benchmark evaluation of core tracking components.

Conclusion: Advancing Football Analytics Workflow

The project successfully validates an end-to-end offline analytics framework for professional football.

- **Tactical Foundation:** Proved that object detection and tracking serves as the critical base for tactical analytics and heuristic event generation.
- **Sequence Modeling Utility:** Integration of temporal sequence modeling demonstrates that engineered temporal features are highly effective for modeling rare or structurally defined football events.
- **Future Research:** Establishes a reproducible and data-driven foundation for advanced predictive modeling in human competition.